

國立陽明交通大學
智慧計算與科技研究所
碩士論文

Institute of Computational Intelligence
National Yang Ming Chiao Tung University
Master Thesis

迭代式組合零樣本影像生成

Iterative Compositional Zero-Shot Generation

研究生：吳佩珊 (Pei-Shan Wu)
指導教授：馬清文 (Ching-Wen Ma)

中華民國 一一五年七月
July 2026

迭代式組合零樣本影像生成
Iterative Compositional Zero-Shot Generation

研 究 生：吳佩珊
指導教授：馬清文

Student: Pei-Shan Wu
Advisor: Dr. Ching-Wen Ma

國立陽明交通大學
智慧計算與科技研究所
碩士論文

A Thesis
Submitted to Institute of Computational Intelligence
College of Science
National Yang Ming Chiao Tung University
in partial Fulfilment of the Requirements
for the Degree of
Master
in
Artificial Intelligence

July 2026

Taiwan, Republic of China

中華民國 一一五年七月

Acknowledgement

完成這篇論文，象徵著我的碩士生涯也即將畫下句點。回首這段求學歷程，從研究方向的摸索、實驗的反覆嘗試，到論文的撰寫與修改，每一步都充滿挑戰，也讓我在過程中累積了許多寶貴的經驗。在此，我想向一路陪伴、支持與幫助我的師長、同學及家人，表達由衷的謝意。

首先，感謝我的指導教授馬清文教授，在研究期間給予我指導與建議，使我能順利完成論文，也讓我在研究能力與思考方式上都有所成長。此外，也感謝各位口試委員撥冗審閱論文，並提供許多寶貴的建議，使本論文更加完善。

感謝實驗室的夥伴們，特別是士軒、秋賢、政霖以及 David，在研究與生活中彼此交流、互相幫助，也因為有你們的陪伴，讓碩士生活留下許多珍貴的回憶。

最後，我要感謝我的家人。謝謝你們一路以來始終支持、鼓勵與包容我，即使這段求學歷程比預期更長，你們依然陪伴著我，成為我最堅強的後盾，讓我能夠堅持完成碩士學業。

這段碩士生活雖然充滿挑戰，卻也讓我學會了堅持與成長。謹以此文，向所有曾經幫助、鼓勵與支持我的人，獻上最誠摯的感謝。

迭代式組合零樣本影像生成

學生：吳佩珊

指導教授：馬清文 教授

國立陽明交通大學 智慧計算與科技研究所

摘要

近年來，擴散模型於影像生成任務中展現出優異的生成品質與條件控制能力，其中 Diffusion Transformer (DiT) 因其以 Transformer 為核心架構而受到廣泛關注。然而，傳統 class-conditional DiT 多以單一標籤作為條件，難以有效表示由多個語意因素所組成的複合條件，因而限制了模型對未見組合類別的生成能力。為了解決此問題，本研究提出一種 Compositional Conditional Diffusion Transformer，將原本的單一類別條件分解為多個語意因素，並以 attribute-object 的組合形式作為條件輸入，使模型能夠學習更具組合性的條件表示，進而生成訓練資料中未出現的影像類別。

雖然 compositional conditioning 能夠提升模型對未見組合的泛化能力，但生成結果仍可能包含不完全符合條件的樣本。為了進一步提升生成影像的正確性，本研究引入 human-in-the-loop 的迭代式再訓練機制，透過人工篩選生成結果中的正確樣本，並將其逐步加入訓練資料中，以改善後續生成品質。然而，人工挑選過程耗時且成本高，當可用的人工驗證資源有限時，新加入資料的類別分布可能變得不平衡，進而影響模型的訓練穩定性與整體表現。

針對上述問題，本文進一步提出 modified balanced sampling 策略，以可調式參數平滑控制不同類別的取樣機率，在維持少數類別可見性的同時，避免過度平衡所造成的過擬合或整體資料分布扭曲。實驗結果顯示，所提出的方法能有效提升未見組合類別的生成能力，改善 human-in-the-loop 迭代再訓練下的樣本正確性，並在資料不平衡的情境下維持較佳的生成多樣性與訓練穩定性。

關鍵詞: 擴散模型、Diffusion Transformer、組合式條件生成、人機協作、迭代再訓練、平衡取樣、影像生成

Iterative Compositional Zero-Shot Generation

Student: Pei-Shan Wu

Advisor: Dr. Ching-Wen Ma

Institute of Computational Intelligence
National Yang Ming Chiao Tung University

Abstract

Recent advances have shown that diffusion models demonstrate strong performance in generating high-quality images, and the Diffusion Transformer (DiT) has further improved generation quality through a Transformer-based architecture. However, conventional class-conditional DiT typically relies on a single label as the conditioning signal, which limits its ability to model compositional semantics and generalize to unseen category combinations. To address this limitation, we propose a Compositional Conditional Diffusion Transformer that decomposes each condition into an attribute-object pair, enabling the model to generate unseen image categories in a more flexible and compositional manner.

To further improve sample correctness, we introduce a human-in-the-loop iterative retraining framework, where manually verified correct samples are progressively added back into the training set. Since manual selection is time-consuming and labor-intensive, the updated data may become imbalanced across classes. Therefore, we propose a modified balanced sampling strategy to control class-wise sampling probabilities and maintain training stability and generation diversity. Experimental results show that the proposed method improves the generation of unseen compositional categories while preserving better sample quality under iterative retraining.

Keyword: diffusion models, Diffusion Transformer, compositional conditional generation, human-in-the-loop, iterative retraining, balanced sampling, image generation

Table of Contents

摘要	i
Abstract	ii
Table of Contents	iii
List of Figures	v
List of Tables	vi
1 Introduction	1
1.1 Background	1
1.2 Research motivation	2
1.2.1 Compositional Conditional DiT for Unseen Combinations	2
1.2.2 Human-in-the-Loop Iterative Retraining	3
1.3 Contribution	4
2 Related Work	5
2.1 Iterative Retraining	5
2.2 Imbalance data technology	5
2.3 Compositional Zero-Shot Learning & generation	7
2.3.1 Compositional Zero-Shot Learning	7
2.3.2 Compositional Conditional Diffusion Model	7
2.4 Diffusion Transformer	8
3 Method	10
3.1 Sampling Strategy	10
3.2 Compositional Diffusion Transformer Model	11
3.2.1 Latent Diffusion Framework	12
3.2.2 Compositional Condition Representation	13
3.2.3 Condition Embedding and Fusion	14
3.2.4 Transformer-Based Denoising Network	15

3.2.5	Training Objective	16
3.2.6	Classifier-Free Guidance	17
3.2.7	Conditional Image Generation	17
3.3	Data diversity evaluation	18
3.4	Classifier-based Evaluation	19
4	Experiments and Results	21
4.1	Dataset	21
4.2	Experiment setup	22
4.2.1	Compositional Diffusion Transformer	22
4.2.2	Classifier-based Evaluation	25
4.3	Evaluation Metrics	26
4.4	Experiment results	27
4.4.1	Quantitative Analysis of Iterative Training on CelebA Dataset	27
4.4.2	Qualitative Analysis of Iterative Training on CelebA Dataset	28
4.4.3	Results of Image Generation	31
4.4.4	Classifier-based Evaluation Results	32
5	Conclusion	34
	References	35

List of Figures

3.1	Overview of the proposed compositional diffusion transformer in latent space. The input image is encoded into a latent representation, denoised under the compositional condition, and decoded back into the image space.	13
3.2	Illustration of the compositional condition fusion strategy. The attribute and object embeddings are first concatenated and passed through an MLP, and the resulting condition representation is added to the timestep embedding before being injected into the DiT block.	15
3.3	LPIPS-based diversity evaluation for generated images.	19
3.4	Overview of the classifier-based evaluation pipeline.	20
4.1	Experimental setup on the CelebA dataset. The matrix on the left shows the training set configuration where the “Gray Hair Female” class is deliberately withheld (marked as unseen). The right panel shows the target generation goal of synthesizing high-quality images for this missing category.	21
4.2	Visual analysis of latent space distributions. (a) In Iteration 2, $\gamma = 0.7$ shows increased concentration. (b) In Iteration 3, the extreme contrast between $\gamma = 0$ (Concentrated) and $\gamma = 1$ (Dispersed) demonstrates the relationship between LPIPS scores and visual diversity.	30
4.3	Visual comparison of generated “Gray Hair Female” samples. By strategically reducing marginal diversity through iterative refinement, the framework significantly enhances generation accuracy while maintaining sufficient visual variety for downstream utility.	32

List of Tables

4.1	Initial sample distribution of the training dataset D_1 across gender and hair color categories.	22
4.2	Experimental setup and model configuration.	24
4.3	Training and sampling settings.	25
4.4	Quantitative results: Number of correct images and Accuracy (%) across iterations and γ values.	28
4.5	Quantitative results: LPIPS scores across iterations and γ values.	28
4.6	Classifier-based evaluation results under different sampling strategies across Iteration 2 and Iteration 3. The test set consists of 1000 Gray_Hair Female images. The final composition accuracy is computed by requiring both the attribute classifier and the object classifier to predict correctly.	33

Chapter 1

Introduction

1.1 Background

In computer vision, image generation models have become an important research topic due to their ability to synthesize diverse and realistic images from different types of conditions, such as text, labels, or input images [1, 2]. These models have been widely applied to style transfer, image inpainting, enhancement, and data synthesis [3, 4, 5]. Among the representative approaches, variational autoencoders (VAEs) [6], generative adversarial networks (GANs) [7], and diffusion models [8, 9] each offer different strengths and limitations. VAEs typically exhibit stable training dynamics, but the resulting images tend to be blurry with limited fine-grained details. In contrast, GANs are capable of generating sharper and more realistic outputs, but they often encounter issues such as training instability and mode collapse [10]. In recent years, diffusion models have become the dominant paradigm for image synthesis, achieving high-quality and diverse generation through an iterative denoising procedure and surpassing GANs on multiple benchmarks [11].

However, traditional diffusion architectures are still commonly built upon the U-Net backbone [12, 9], which is effective at capturing local image structures but less capable of modeling long-range dependencies and complex global context [13]. This limitation becomes more apparent when the task requires richer semantic consistency and better diversity in generated images. In contrast, Transformer architectures [13] have shown outstanding results across natural language understanding and image analysis tasks [14]. These models excel at capturing global relationships across an entire image, making them a promising alternative for image generation. Based on this motivation, our research adopts the Diffusion Transformer (DiT) [15] as the core framework to improve image generation quality and diversity.

1.2 Research motivation

1.2.1 Compositional Conditional DiT for Unseen Combinations

Traditional class-conditional Diffusion Transformers (DiT) [15] typically model each visual category as a single atomic label. This formulation inherently limits how well the model can generalize to compositional semantic structures, as it treats related concepts as entirely independent entities. To overcome this limitation, we propose replacing atomic single-label conditioning with a *compositional conditional representation*. By decomposing image conditions into multiple semantic factors—specifically attribute-object pairs—the model can learn to disentangle these concepts [16, 17]. This design enables the model to perform combinatorial generalization, synthesizing unseen combinations that were absent during the initial training phase.

This compositional zero-shot image generation and selection framework is particularly valuable in data-scarce domains such as medical imaging. One potential application is in dermatology, where the model could generate skin lesion images that are difficult or time-consuming to collect in practice. Dermatologists could then filter, select, and annotate the generated images to construct task-specific classifiers for clinical decision support. This envisioned human-in-the-loop pipeline illustrates how our method can reduce the burden of data collection while enabling the creation of balanced and representative datasets for rare or underrepresented conditions.

Our approach differs from prior DiT and conditional diffusion models that rely on a single class label or fixed conditioning format. While existing works have explored compositional generation in GANs or other architectures, few have investigated compositional conditioning within the DiT framework under severe class imbalance, nor have they integrated expert-in-the-loop selection for downstream classifier construction. By explicitly modeling attribute and object factors and targeting unseen combinations in this setting, our method extends prior conditional diffusion approaches to more flexible and data-efficient compositional generation, with direct applicability to real-world medical imaging workflows.

1.2.2 Human-in-the-Loop Iterative Retraining

While compositional conditioning provides the capacity for zero-shot synthesis, the reliability of generated samples remains a significant challenge. In complex generative tasks, models often produce “hallucinations” or semantic inconsistencies where the output fails to accurately reflect the target condition [18]. To bridge this quality gap, we introduce a *Human-in-the-Loop (HITL)* iterative retraining procedure. By incorporating manual inspection into the feedback loop, we ensure that only factually correct synthetic samples are used to augment the training set. Recent studies suggest that such curated data mechanisms can act as an implicit reward signal, effectively reducing model collapse and improving alignment with human preferences [19, 20].

However, the inclusion of HITL introduces a practical bottleneck: human verification is inherently time-consuming and labor-intensive. In real-world scenarios with limited annotation budgets, this labor constraint often results in a sparse and uneven distribution of verified samples. Consequently, the augmented dataset may exhibit severe class imbalances, which have been shown to hinder the learning stability of generative models [21].

This iterative retraining framework is especially relevant in scenarios with limited expert availability. A small number of experts can iteratively review and select generated samples, enabling high-quality data creation under tight annotation budgets and modest human effort. However, this process naturally leads to an imbalanced augmented dataset, which motivates the need for a tailored sampling strategy during training.

Modified Balanced Sampling

To mitigate the distributional shift and class imbalance introduced by the iterative retraining process, we further propose a *modified balanced sampling* strategy. Standard uniform sampling is often insufficient in this context, as it tends to favor majority classes and neglect the rare synthetic samples critical for learning unseen combinations [22]. Our approach employs a **smoothing parameter**, γ , to assign dynamic sampling probabilities based on class frequency. This mechanism ensures that underrepresented classes receive sufficient attention during train-

ing without causing the model to lose its representation of the dominant classes.

Ultimately, our research is motivated by the need for a unified framework that harmonizes compositional generation, human-guided refinement, and balanced optimization. By introducing the parameter γ to regulate the intensity of re-balancing, we aim to build a robust system that can reliably synthesize unseen attribute combinations while maintaining a stable and balanced learning trajectory throughout the iterative process.

1.3 Contribution

The primary contributions of this research, which bridge prior laboratory concepts with modern transformer architectures and original optimization strategies, are summarized as follows:

- **Transitioning Compositional Conditioning to DiT:** Building upon the attribute-object factorization concept previously developed in our laboratory for U-Net architectures, we successfully transition and validate this paradigm within the **Diffusion Transformer (DiT)** framework. This research provides a systematic evaluation of how compositional labels enable zero-shot synthesis in transformer-based diffusion models, expanding the applicability of this concept beyond traditional CNN backbones.
- **Iterative Generation and Refinement Framework:** We develop an original iterative data generation and filtering framework that incorporates human-in-the-loop (HITL) feedback. By utilizing manual selection to mitigate semantic "hallucinations" in unseen combinations, the framework progressively improves the fidelity and semantic accuracy of zero-shot synthesis over multiple training cycles.
- **Original γ -based Modified Balanced Sampling:** We propose an original sampling weight adjustment strategy based on a smoothing parameter γ . This strategy is specifically designed to handle the class imbalance and sparsity introduced by the HITL process. It provides a robust mechanism to stabilize the training trajectory, effectively balancing the trade-off between improving semantic correctness and preserving perceptual diversity (LPIPS).

Chapter 2

Related Work

2.1 Iterative Retraining

Iterative retraining, often referred to as self-consuming generative modeling, studies the learning dynamics that arise when a model’s synthetic outputs are fed back into its training pipeline. A major concern in this area is ”model collapse,” in which the model gradually forgets rare modes and converges to a narrow, low-diversity distribution.

This phenomenon has been shown to occur when recursive training is performed without sufficient real-world data, leading to irreversible quality degradation [20]. Further theoretical work indicates that the stability of such training loops is highly sensitive to the initial model quality and the ratio between clean and synthetic data [21].

To mitigate these risks, recent research emphasizes the necessity of data intervention. It has been demonstrated that curated data mechanisms can act as an implicit reward signal, reducing collapse and improving alignment with human preferences [19]. This suggests that the success of iterative training loops is not merely a function of data volume, but depends heavily on the sampling and filtering strategies used to select high-quality synthetic samples. As such, the balance between newly generated synthetic data and the original data distribution becomes a critical design factor, which we investigate in the context of sampling techniques.

2.2 Imbalance data technology

Data imbalance, particularly in the form of long-tailed distributions, remains a fundamental challenge in machine learning. In such scenarios, majority classes dominate the training set, while minority classes—often containing the most critical information for generalization—are

underrepresented [23]. This imbalance causes models to exhibit a strong bias toward majority classes, leading to poor performance on rare or unseen attribute combinations.

Conventional solutions to this problem typically fall into two categories: re-sampling and re-weighting. Re-sampling strategies adjust the class distribution by either duplicating minority class samples or removing majority class samples to achieve a more balanced training set [24]. However, aggressive over-sampling can lead to over-fitting on minority samples, while under-sampling may discard valuable information from majority classes. Alternatively, re-weighting strategies adjust the loss function by assigning higher weights to minority samples. Class-balanced loss methods have been proposed to better reflect the marginal benefit of adding new data points, and bilateral-branch designs have also been explored to decouple feature learning from classifier balancing.

In the context of generative models, data imbalance often causes *mode collapse*, in which the model fails to learn the complete data distribution and instead generates samples from only a limited set of high-frequency modes [25]. Recent research in diffusion models has shown that while these architectures are more robust than GANs, they are still sensitive to the training data distribution. In imbalanced generative tasks, standard uniform sampling can significantly reduce the fidelity of minority class generation.

While existing sampling strategies often pursue a strict uniform distribution, our work identifies a critical trade-off between semantic accuracy and perceptual diversity. Extreme class balancing can sometimes harm the model’s ability to represent the global data manifold. Therefore, we investigate a *Modified Balanced Resampling* strategy. By introducing a tunable parameter γ , we enable a smoothed transition between uniform and balanced sampling, allowing for a strategic refinement of the generation space that is particularly effective in iterative human-in-the-loop training environments.

2.3 Compositional Zero-Shot Learning & generation

2.3.1 Compositional Zero-Shot Learning

Compositional Zero-Shot Learning (CZSL) aims to achieve *combinatorial generalization* by learning to recognize or synthesize novel combinations of previously seen semantic primitives, such as attributes and objects [17]. The fundamental challenge of CZSL lies in the disentanglement of these primitives from their original training contexts [26]. For example, a model may be trained on seen pairs such as [dry] $\times$ [dog] and [wet] $\times$ [cat]; the ultimate objective is to enable the model to compose the unseen pairs [wet] $\times$ [dog] or [dry] $\times$ [cat] by effectively decoupling the attribute from its associated object.

In recent years, the scope of this research area has expanded from purely discriminative tasks to **Compositional Zero-Shot Generation (CZSG)**. The goal of CZSG is to synthesize high-fidelity images for these novel combinations by modeling the conditional distribution $P(\text{Image} \mid \text{Attribute, Object})$ [27]. Reliably bridging this semantic gap requires a robust conditioning mechanism that can accurately perform *attribute binding*—the process of correctly associating specific attributes with target objects in a zero-shot setting [26, 27]. This challenge of semantic composition and attribute binding motivates the specialized generative architectures and conditioning strategies that will be explored in the following sections.

2.3.2 Compositional Conditional Diffusion Model

The Compositional Conditional Diffusion Model (CCDM) [28] is proposed to improve the compositionality of diffusion models for unseen image generation. The key idea is to represent a condition as a composition of multiple semantic components rather than a single atomic label, enabling the model to synthesize image combinations that are not explicitly present in the training data. This compositional formulation is particularly useful for zero-shot image generation, where the target condition may be formed by attribute-object pairs that were not observed during training.

By guiding the diffusion process with compositional class labels, CCDM demonstrates that smaller diffusion models can still achieve controllable zero-shot generation without relying on large-scale foundation models. The study further shows that compositional class labels provide a practical way to extend generation capability beyond the training distribution, highlighting the potential of factorized conditional representations for domain-specific image synthesis. This work serves as an important motivation for our research, as it suggests that compositional conditioning can effectively improve unseen image generation while remaining lightweight and domain adaptable.

2.4 Diffusion Transformer

Addressing the inherent limitations of convolutional backbones in capturing truly global interactions within high-dimensional data, recent research has pivotally shifted towards integrating Transformer architectures [13] into the diffusion framework. Transformers, first developed for natural language processing tasks like machine translation and language modeling, have since demonstrated remarkable versatility and achieved leading success across a diverse array of domains, including prominently in computer vision [14]. Their fundamental strength lies in the self-attention mechanism. Unlike convolutional operations that process information locally, self-attention allows the model to compute the representation of each element or token in a sequence by dynamically weighing its interaction with all other elements in the same sequence. This is achieved by projecting each input token into query, key, and value vectors, and then computing attention scores based on the dot products of queries and keys, which subsequently modulate the values. This mechanism effectively enables every token to "attend" to all other tokens, irrespective of their spatial or sequential distance, thereby creating a global receptive field from the outset and making Transformers exceptionally adept at modeling long-range dependencies and global contextual relationships.

Pioneering this architectural integration within diffusion models, Peebles and Xie [15] introduced the Diffusion Transformer (DiT). The DiT paradigm fundamentally re-envisioned the

denoising network by replacing the conventional U-Net-like convolutional backbone with a pure Transformer architecture. The operational pipeline of a DiT typically begins by partitioning the input image—or, in many efficient variants, its corresponding latent representation obtained from a pre-trained autoencoder, following an approach similar to that of Latent Diffusion Models (LDMs)—into a sequence of non-overlapping, flattened patches. The patches are then transformed into patch embeddings through a linear projection. To retain spatial information, positional embeddings, often learned or sinusoidal, are added to these patch embeddings. The resulting sequence of patch embeddings, along with a conditioning signal that represents the current diffusion timestep t as well as potentially other class or text conditions, is then fed through a stack of standard Transformer blocks. Each Transformer block is typically composed of a multi-head self-attention module followed by a feed-forward network in the form of a multilayer perceptron, with Layer Normalization applied before each sub-layer and residual connections around them. The multi-head self-attention layer is crucial as it allows the model to simultaneously attend to information from different representation subspaces across different positions. Through these stacked blocks, self-attention mechanisms capture intricate interactions and dependencies across all patch embeddings, empowering the model to learn complex, global relationships and spatial arrangements throughout the iterative denoising process.

Chapter 3

Method

3.1 Sampling Strategy

To handle label imbalance during training, we propose a modified balanced sampling strategy that assigns higher sampling probability to underrepresented classes while preserving flexibility through a smoothing parameter γ . Given a dataset $\mathcal{D}$ with class labels L , we first compute the frequency N_i of each class. The unnormalized sampling probability for sample i is defined as

$$p_i = \frac{1}{N_i^\gamma}. \quad (3.1)$$

The final sampling distribution is then obtained by normalizing these probabilities:

$$P_i = \frac{p_i}{\sum_j p_j}. \quad (3.2)$$

During each training epoch, samples are drawn according to this probability distribution using weighted random sampling with replacement. This enables minority classes to be selected more frequently without completely discarding information from majority classes. As a result, the training process remains stochastic while reducing the impact of class imbalance.

The smoothing parameter γ controls the degree of balancing. When $\gamma = 1$, the method reduces to standard balanced sampling, which gives the strongest emphasis to rare classes. When $0 < \gamma < 1$, the sampling becomes smoother and provides a compromise between class balance and diversity preservation. When $\gamma = 0$, the method degenerates to uniform sampling, where all samples are selected with equal probability. This design allows the proposed strategy to adapt to different imbalance levels and training requirements.

In our implementation, the sampling probabilities are recomputed from the class frequen-

cies in the dataset and used to generate the training indices for each epoch. This balanced yet stochastic sampling mechanism is particularly useful for iterative retraining in compositional conditional image generation.

Algorithm 3.1 Modified Balanced Sampler

Require: Dataset $\mathcal{D}$ of size N , labels L , smoothing parameter γ , seed s , epoch e

Ensure: An iterator of N sample indices

- 1: Count class frequency for each unique class in L
 - 2: Initialize probability list $P \leftarrow \emptyset$
 - 3: **for** each sample i with class label c **do**
 - 4: $p_i \leftarrow 1/(N_c^\gamma)$
 - 5: Append p_i to P
 - 6: **end for**
 - 7: Normalize probabilities: $P \leftarrow P / \sum(P)$
 - 8: Set random seed for the current epoch: $seed \leftarrow s + e$
 - 9: Sample N indices from $\{0, \dots, N - 1\}$ with replacement according to P
 - 10: Shuffle sampled indices if needed
 - 11: **return** the sampled indices
-

3.2 Compositional Diffusion Transformer Model

This section describes the proposed Compositional Diffusion Transformer, which extends the class-conditional Diffusion Transformer (DiT) by decomposing the condition into multiple semantic factors. Instead of treating each visual class as an indivisible category, the proposed model represents an image condition as the composition of an attribute and an object. This design enables the model to learn factorized conditional representations and generate images from explicit attribute-object combinations.

The overall training procedure, which integrates compositional conditioning, human-in-the-loop iterative retraining, and modified balanced sampling, is summarized in Algorithm 3.2.

Algorithm 3.2 Compositional Conditional DiT with HITL Iterative Retraining and Modified Balanced Sampling

Require: Initial dataset $\mathcal{D}_0$ (without target unseen class), smoothing parameter γ , number of iterations T

Ensure: Final refined model f_{θ_T}

- 1: **Phase 1: Initial Zero-Shot Training**
 - 2: Train initial model f_{θ_0} on $\mathcal{D}_0$ using compositional conditioning
 - 3: **Phase 2: Human-in-the-Loop Iterative Refinement**
 - 4: **for** $t = 1$ **to** T **do**
 - 5: Generate candidate images for unseen attribute–object combinations using $f_{\theta_{t-1}}$
 - 6: **Human Selection:** Evaluators manually select a subset $\mathcal{S}_t$ of correct, high-quality samples
 - 7: Update dataset: $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \mathcal{S}_t$
 - 8: **Re-balancing:** Recompute sampling weights based on class frequencies in $\mathcal{D}_t$ and parameter γ (refer to Alg. 3.1)
 - 9: Retrain/Fine-tune model to obtain f_{θ_t} using the updated modified balanced sampling
 - 10: **end for**
 - 11: **return** f_{θ_T}
-

3.2.1 Latent Diffusion Framework

The proposed method follows a latent diffusion framework. Given an input image x , a pretrained variational autoencoder (VAE) encoder $\mathcal{E}(\cdot)$ is used to project the image into a latent representation:

$$z_0 = 0.18215 \cdot \mathcal{E}(x).$$

Training is performed in the latent space, as opposed to the raw pixel space. This significantly reduces the spatial resolution of the denoising input and improves computational efficiency. For example, an image with resolution 128×128 is encoded into a latent representation with spatial size 16×16 and four channels.

At each training iteration, a diffusion timestep t is sampled, and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ is injected into the clean latent z_0 :

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad (3.3)$$

where $\bar{\alpha}_t$ denotes the cumulative product of the noise schedule. In this work, a linear noise schedule with 1000 diffusion steps is used.

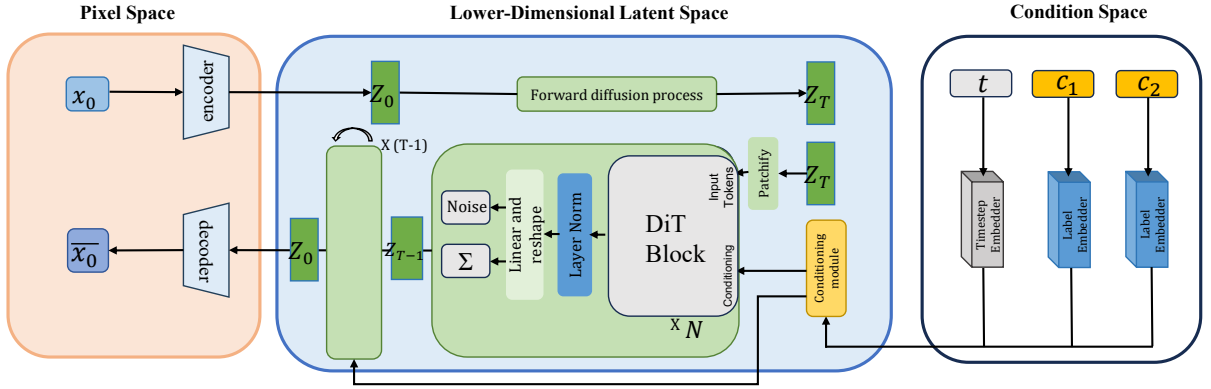


Figure 3.1: Overview of the proposed compositional diffusion transformer in latent space. The input image is encoded into a latent representation, denoised under the compositional condition, and decoded back into the image space.

3.2.2 Compositional Condition Representation

Unlike conventional class-conditional DiT, which uses a single class label as the condition, the proposed model decomposes each condition into two semantic components:

$$(c_{\text{atr}}, c_{\text{obj}}),$$

where c_{atr} denotes the attribute condition and c_{obj} denotes the object condition. For the CelebA dataset, the attribute condition corresponds to hair color, such as Brown Hair, Blond Hair, Gray Hair, and Black Hair, while the object condition corresponds to gender, such as Male and Female.

Therefore, a class such as “Gray Hair Female” is represented as the composition of two

separate factors:

$$\text{Gray Hair Female} = (\text{Gray Hair}, \text{Female}).$$

During data loading, the folder name of each image is split into its attribute and object components. The two labels are then mapped to integer indices using two independent dictionaries:

$$c_{\text{atr}} \mapsto y_{\text{atr}}, \quad c_{\text{obj}} \mapsto y_{\text{obj}}.$$

These two indices are used as conditional inputs to the denoising network.

3.2.3 Condition Embedding and Fusion

To encode the compositional condition, the model uses two independent learnable embedding layers. The attribute and object labels are embedded separately:

$$e_{\text{atr}} = E_{\text{atr}}(y_{\text{atr}}), \quad e_{\text{obj}} = E_{\text{obj}}(y_{\text{obj}}),$$

where E_{atr} and E_{obj} are the attribute and object embedding tables, respectively.

The two condition embeddings are first concatenated and then projected into the hidden space by a multilayer perceptron:

$$e_{\text{cond}} = \text{MLP}_{\text{cond}}([e_{\text{atr}}; e_{\text{obj}}]).$$

A sinusoidal positional encoding is first applied to the diffusion timestep t , followed by a multilayer perceptron for embedding:

$$e_t = E_t(t).$$

The final conditioning vector is obtained by element-wise addition of the transformed condition embedding and the timestep embedding:

$$c = e_t + e_{\text{cond}}.$$

This design first concatenates the attribute and object embeddings, projects the fused condition through an MLP, and then adds the resulting representation to the diffusion timestep embedding before feeding it into the DiT block. Instead of learning a separate embedding for every possible attribute-object pair, the model learns the two factors independently and combines them dynamically during both training and inference.

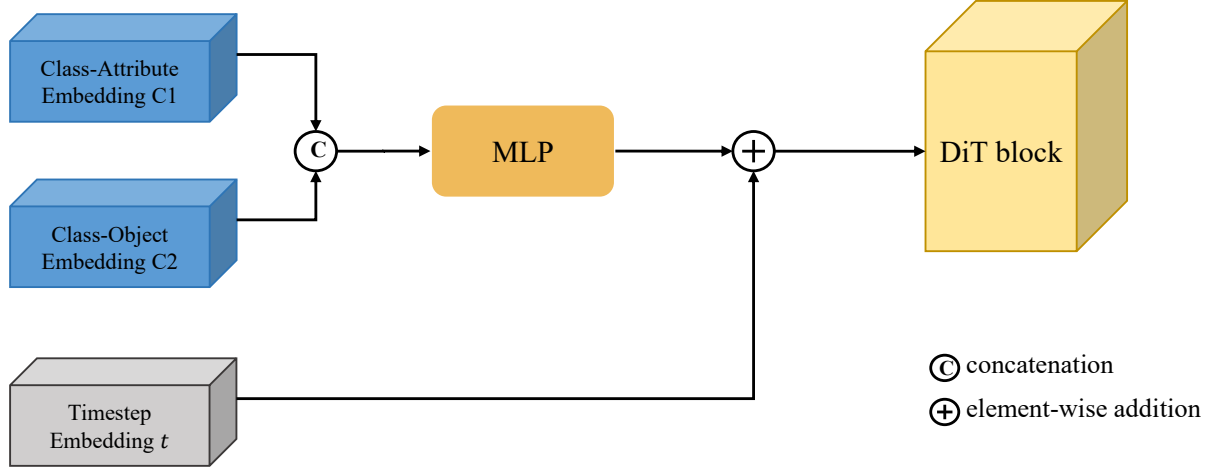


Figure 3.2: Illustration of the compositional condition fusion strategy. The attribute and object embeddings are first concatenated and passed through an MLP, and the resulting condition representation is added to the timestep embedding before being injected into the DiT block.

3.2.4 Transformer-Based Denoising Network

The noisy latent z_t is first divided into non-overlapping patches and projected into a sequence of latent tokens:

$$h_0 = \text{PatchEmbed}(z_t) + p,$$

where p is a fixed two-dimensional sinusoidal positional embedding. A succession of Transformer blocks then operates on the token sequence to extract complex features.

Each Transformer block follows the DiT design and uses adaptive layer normalization with zero initialization, known as adaLN-Zero. The compositional conditioning vector c is used to generate modulation parameters for both the multi-head self-attention branch and the feed-forward branch:

$$(\Delta_{\text{msa}}, \Gamma_{\text{msa}}, G_{\text{msa}}, \Delta_{\text{mlp}}, \Gamma_{\text{mlp}}, G_{\text{mlp}}) = \text{MLP}_{\text{adaLN}}(c).$$

Here, Δ and Γ denote shift and scale parameters, while G denotes a gating parameter. The modulation operation is defined as:

$$\text{Modulate}(h, \Delta, \Gamma) = h \odot (1 + \Gamma) + \Delta.$$

The self-attention branch is computed as:

$$h' = h + G_{\text{msa}} \odot \text{MSA}(\text{Modulate}(\text{LN}(h), \Delta_{\text{msa}}, \Gamma_{\text{msa}})),$$

and the feed-forward branch is computed as:

$$h_{\text{out}} = h' + G_{\text{mlp}} \odot \text{MLP}(\text{Modulate}(\text{LN}(h'), \Delta_{\text{mlp}}, \Gamma_{\text{mlp}})).$$

Through this design, the same compositional condition vector is injected into every Transformer block. The denoising process is therefore guided jointly by the diffusion timestep, the attribute condition, and the object condition.

3.2.5 Training Objective

The training objective of the denoising architecture is to estimate the specific Gaussian noise that corrupts the latent representation. Given the noisy latent z_t , timestep t , attribute label y_{atr} , and object label y_{obj} , the model predicts:

$$f_{\theta}(z_t, t, y_{\text{atr}}, y_{\text{obj}}) = (\hat{\epsilon}_{\theta}, \hat{\Sigma}_{\theta}).$$

The primary denoising loss is the mean squared error between the true noise ϵ and the predicted noise:

$$\mathcal{L}_{\text{mse}} = \mathbb{E}_{z_0, t, \epsilon} [\|\epsilon - \hat{\epsilon}_{\theta}(z_t, t, y_{\text{atr}}, y_{\text{obj}})\|_2^2]. \quad (3.4)$$

Because the model also predicts the reverse-process variance, an additional variational bound

term is included:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda_{\text{vb}} \mathcal{L}_{\text{vb}}. \quad (3.5)$$

The implementation uses a rescaled MSE objective, where the variational bound term is rescaled according to the number of diffusion timesteps. This stabilizes training when both the denoising direction and the reverse-process variance are learned.

3.2.6 Classifier-Free Guidance

Classifier-free guidance is used to improve conditional generation quality. The condition embedding layers support label dropout, where a condition label may be replaced by a special null label during training. This allows the same network to learn both conditional and unconditional denoising.

During inference, the conditional and unconditional predictions are combined as:

$$\hat{\epsilon}_{\text{cfg}} = \hat{\epsilon}_{\text{uncond}} + s (\hat{\epsilon}_{\text{cond}} - \hat{\epsilon}_{\text{uncond}}), \quad (3.6)$$

where s is the guidance scale. A larger s strengthens the influence of the given attribute-object condition during generation.

3.2.7 Conditional Image Generation

During inference, image generation starts from a Gaussian latent variable:

$$z_T \sim \mathcal{N}(0, I).$$

Given a desired attribute-object pair $(y_{\text{atr}}, y_{\text{obj}})$, the reverse diffusion process iteratively denoises z_T into z_0 under the specified compositional condition. The final latent is then decoded back into image space using the VAE decoder:

$$\hat{x} = \mathcal{D}(z_0/0.18215), \quad (3.7)$$

where $\mathcal{D}(\cdot)$ denotes the VAE decoder.

This process enables the model to generate images from explicit compositional conditions. For example, by specifying the attribute condition “Gray Hair” and the object condition “Female”, the model is guided to synthesize images corresponding to the composition “Gray Hair Female”.

3.3 Data diversity evaluation

To examine the utility of the generated compositional images, we adopt the Learned Perceptual Image Patch Similarity (LPIPS) criterion as a perceptual distance measure. Unlike pixel-wise measures, LPIPS compares images in deep feature space and is therefore more consistent with human perception. In our setting, this metric is used to quantify how diverse the generated samples are under different condition combinations.

As illustrated in Fig. 3.3, given two images x and x' , we first extract multi-level feature representations from a pretrained CNN backbone, such as VGG. Let $\{f_l(x)\}_{l=1}^L$ and $\{f_l(x')\}_{l=1}^L$ denote the feature maps from layer l . Each layer-wise feature is channel-wise normalized, and the perceptual distance at layer l is computed using the squared ℓ_2 distance:

$$d_l = \left\| \hat{f}_l(x) - \hat{f}_l(x') \right\|_2^2.$$

The final LPIPS score is then obtained by aggregating the distances across layers with learned weights:

$$\text{LPIPS}(x, x') = \sum_{l=1}^L w_l d_l.$$

A larger LPIPS value indicates greater perceptual difference between two images, and thus higher diversity among generated samples.

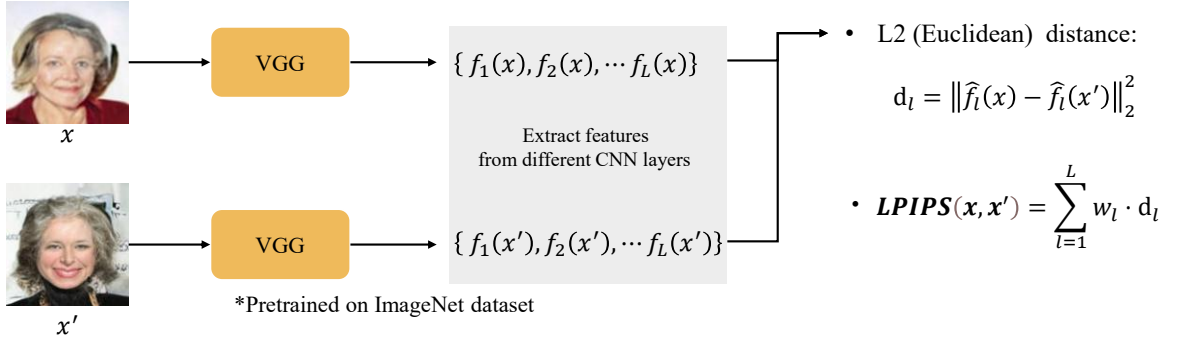


Figure 3.3: LPIPS-based diversity evaluation for generated images.

3.4 Classifier-based Evaluation

To evaluate the usefulness of the generated compositional images, we train two binary classifiers: an attribute classifier and an object classifier. The attribute classifier is used to verify whether the generated sample contains the correct attribute, while the object classifier is used to verify whether the sample contains the correct object. A sample is considered correctly classified only when both classifiers make correct predictions; i.e., the final accuracy is computed by taking the intersection of the two decisions.

We consider a missing-category setting, where one compositional class is removed from the training set and reserved as the test set. This setting simulates a low-data or unseen-composition scenario in which the model must generalize to a category that is not directly observed during training. To improve the classifier’s ability, we augment the original training set with images generated by our compositional diffusion model and compare the performance against training with real data only.

Figure 3.4 illustrates the overall evaluation pipeline. The generated samples are used as additional training data for the missing category, and the final classification performance is measured on the held-out test set. This evaluation allows us to assess whether the generated images can effectively enhance downstream classification performance.

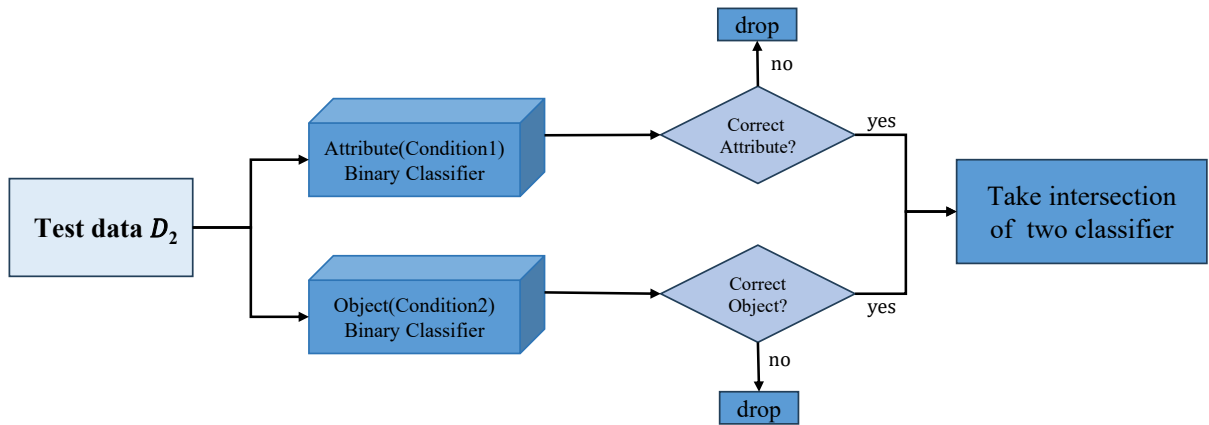


Figure 3.4: Overview of the classifier-based evaluation pipeline.

Chapter 4

Experiments and Results

4.1 Dataset

Validation of the proposed framework’s zero-shot synthesis and compositional generalization abilities is carried out through experiments centered on the **CelebA (CelebFace Attributes Dataset)** [29]. CelebA is a widely-used large-scale face attribute dataset consisting of 202,599 celebrity images, each annotated with 40 binary attribute labels. In our study, we focus on a subset of these attributes to define a controlled experimental environment, specifically utilizing *Gender* as the object condition and *Hair Color* as the attribute condition.

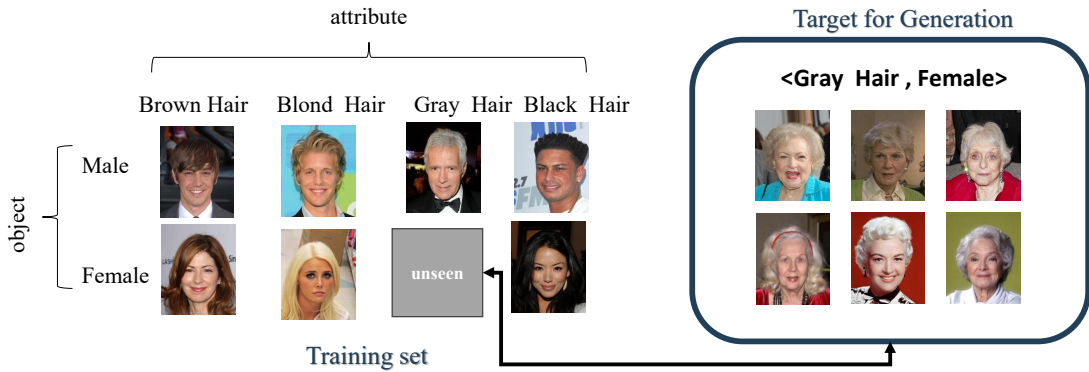


Figure 4.1: Experimental setup on the CelebA dataset. The matrix on the left shows the training set configuration where the “Gray Hair Female” class is deliberately withheld (marked as unseen). The right panel shows the target generation goal of synthesizing high-quality images for this missing category.

We select four representative hair colors—Brown, Blond, Gray, and Black—and combine them with two gender categories—Male and Female—to form a 2×4 compositional matrix. To simulate a zero-shot generation scenario, we deliberately exclude all samples belonging to the **[Gray Hair] $\times$ [Female]** category from the initial training set D_1 . This setup requires the model to learn the decoupled semantic representations of “Gray Hair” and “Female” from other

existing combinations and subsequently synthesize the missing class during inference.

The initial sample distribution of the training dataset D_1 is detailed in Table 4.1. As shown, the target class for generation is initialized with zero samples to represent a deliberate semantic gap for zero-shot evaluation. While the remaining categories follow the natural distribution of the CelebA dataset, our primary focus is on the model’s ability to bridge the specific data vacuum created by the excluded attribute-object pair. This setup serves as a testbed for our iterative refinement process and the *Modified Balanced Sampling* strategy in a controlled yet realistic data environment.

Table 4.1: Initial sample distribution of the training dataset D_1 across gender and hair color categories.

Object (Gender)	Attribute (Hair Color)			
	Brown Hair	Blond Hair	Gray Hair	Black Hair
Male	12,638	1,676	6,644	19,621
Female	20,900	24,358	0 (Unseen)	19,960

By withholding a specific attribute-object pair, we can quantitatively assess the effectiveness of the generative model in synthesizing missing semantic combinations. Any successful generation of high-quality images in the “Gray Hair Female” category demonstrates the model’s capacity for compositional reasoning rather than simple pattern memorization of the training distribution.

4.2 Experiment setup

4.2.1 Compositional Diffusion Transformer

All experiments were conducted using the proposed Compositional Diffusion Transformer. The model was trained in the latent space of a pretrained Stable Diffusion VAE. All input images were resized to 128×128 and normalized to the range $[-1, 1]$ before being encoded into latent representations. The VAE compressed each image into a latent representation with spatial resolution 16×16 and 4 channels.

The diffusion process used a linear noise schedule with 1000 timesteps. The denoising network was based on DiT-XL/2, with a patch size of 2, hidden dimension of 1152, Transformer depth of 28, and 16 attention heads. The compositional condition was represented by two independent factors, namely the attribute condition and the object condition. The two condition embeddings were first concatenated and then projected through an MLP to obtain a unified compositional condition embedding. This embedding was subsequently added to the timestep embedding and used as the conditioning vector for the denoising network.

The model was trained for 200 epochs using the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . During training, an exponential moving average of the model parameters was maintained with a decay rate of 0.9999. Model checkpoints were saved every 50 epochs. During inference, images were generated using 250 sampling steps with classifier-free guidance.

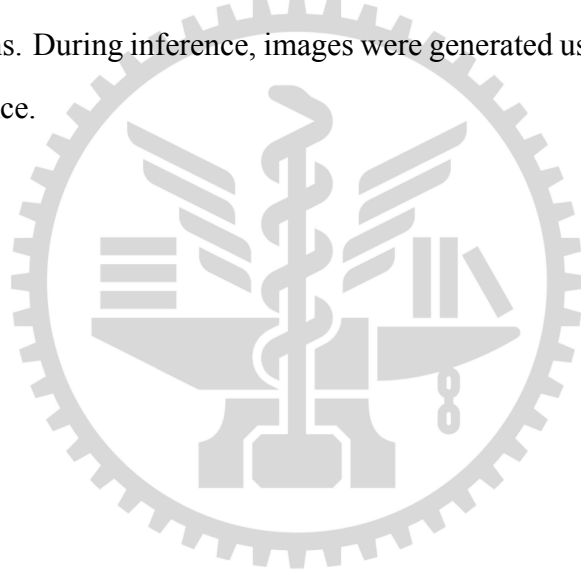


Table 4.2: Experimental setup and model configuration.

Experiment Setup	
Input image resolution	128×128
Image normalization	$[-1, 1]$
Training space	VAE latent space
VAE	Stable Diffusion VAE
Latent resolution	16×16
Latent channels	4
Diffusion Setting	
Diffusion steps	1000
Noise schedule	Linear
Prediction target	Noise ϵ
Learned variance	Yes
Loss type	Rescaled MSE
Model Architecture	
Backbone	DiT-XL/2
Patch size	2
Hidden dimension	1152
Transformer depth	28
Number of attention heads	16
MLP ratio	4.0
Condition injection	adaLN-Zero
Class dropout probability	0.1

Table 4.3: Training and sampling settings.

Training Setting	
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Global batch size	120
Epochs	200
EMA decay	0.9999
Checkpoint interval	50 epochs
Sampling Setting	
Sampling steps	250
Sampling method	DDPM ancestral sampling
Classifier-free guidance	Yes
CFG scale	1.5–2.0
Initial latent distribution	$\mathcal{N}(0, I)$

4.2.2 Classifier-based Evaluation

To evaluate whether the generated images satisfy the target compositional concept, we trained two independent binary classifiers: one for the attribute and one for the object category. Both classifiers were implemented using ResNet-18 without ImageNet pre-trained weights. The final fully connected layer was replaced by a binary classification head consisting of a $512 \rightarrow 256$ linear layer, a ReLU activation, dropout with probability 0.5, and a final $256 \rightarrow 1$ linear layer.

All input images were resized to 128×128 , converted to tensors, and normalized to the range $[-1, 1]$. The attribute and object classifiers were trained separately using their corresponding metadata files. We used binary cross-entropy with logits as the loss function and optimized the models with Adam using a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . The batch size was set to 128, and each classifier was trained for 15 epochs. The best checkpoint for each classifier was selected according to validation accuracy and saved for the final evaluation.

During testing, the trained attribute and object classifiers were applied to the generated test

images. The output logits were passed through a sigmoid function and thresholded at 0.5 to determine whether each image satisfied the target attribute and object condition. An image was considered compositionally correct only when both classifiers predicted positive results. We recorded the number of images classified as attribute-correct, object-correct, and jointly correct, and saved the jointly correct generated samples for further qualitative inspection.

4.3 Evaluation Metrics

To provide a quantitative assessment of the efficacy of our framework,, we define the **Correctness Ratio** as the primary metric. This metric is specifically designed to measure the semantic accuracy of the generated images for unseen attribute combinations.

Correctness Ratio

Due to the difficulty of capturing fine-grained semantic correctness (e.g., ensuring both “Gray Hair” and “Female” features are accurately represented) with automated perceptual metrics, we rely on a human-in-the-loop manual inspection process. For each iteration and sampling configuration, we randomly generate a fixed batch of images targeting the unseen combination. Human evaluators then manually verify whether each image correctly satisfies the specified attribute and object conditions without significant visual artifacts. The **Correctness Ratio** is defined by the following formula:

$$\text{Correctness Ratio} = \frac{\text{number of manually selected correct images}}{\text{number of randomly generated unseen-type images}} \quad (4.1)$$

A higher correctness ratio indicates that the model has successfully learned to bridge the semantic data gap and can reliably synthesize images for combinations not present in the training set. This metric serves as a direct indicator of the effectiveness of our iterative refinement process.

Perceptual Diversity

In addition to semantic accuracy, we monitor the visual diversity of the generated images using the **Learned Perceptual Image Patch Similarity (LPIPS)** [30]. As detailed in the chapter 3.3, LPIPS measures the perceptual distance between deep feature representations of image pairs. By analyzing the interplay between the Correctness Ratio and LPIPS across different training iterations, we can quantitatively assess the strategic trade-off facilitated by our proposed modified balanced sampling strategy.

4.4 Experiment results

4.4.1 Quantitative Analysis of Iterative Training on CelebA Dataset

In this section, we provide a quantitative evaluation of the proposed framework across multiple training iterations. To assess the impact of different sampling intensities, we conduct a comparative study using various configurations of the sampling parameter $\gamma \in \{0, 0.3, 0.5, 0.7, 1\}$.

Specifically, the experimental procedure follows a multi-stage iterative logic:

- For the configurations where $\gamma \in \{0.3, 0.5, 0.7\}$, the training process initiates from **Iteration 2**, utilizing the synthetic output generated from **Iteration 1** under the balanced sampling setting ($\gamma = 1$) as the starting input.
- The training dataset for **Iteration 3** is adaptively constructed by merging the original training set D_1 with the filtered synthetic images generated by each respective γ configuration in **Iteration 2**.

Correctness and Sample Volume

Table 4.4 presents the number of manually selected correct images and the corresponding correctness ratio (Accuracy) for the target unseen category. The results demonstrate a clear upward trend in generation accuracy as the iterations progress. Notably, the proposed modified balanced sampling with $\gamma = 0.7$ achieves the highest correctness ratio of **35.20%** in Iteration 3,

yielding 1,751 high-quality synthetic samples. This confirms that strategically weighting minority classes significantly enhances the model’s ability to synthesize accurate unseen attribute combinations.

Table 4.4: Quantitative results: Number of correct images and Accuracy (%) across iterations and γ values.

Iteration	Random ($\gamma = 0$)	$\gamma = 0.3$	$\gamma = 0.5$	$\gamma = 0.7$	Balanced ($\gamma = 1$)
Iteration 1	473 (9.46%)	—	—	—	622 (12.44%)
Iteration 2	1,085 (21.70%)	1,231 (24.62%)	1,123 (24.16%)	1,132 (22.64%)	1,065 (21.30%)
Iteration 3	1,579 (31.58%)	1,641 (32.82%)	1,588 (31.76%)	1,751 (35.20%)	1,708 (34.16%)

Perceptual Diversity Analysis

To evaluate the visual diversity of the generated images, we report the Learned Perceptual Image Patch Similarity (LPIPS) scores in Table 4.5. Consistent with our research hypothesis, we observe a strategic trade-off between diversity and accuracy.

As γ increases, the LPIPS scores generally decrease or stabilize compared to the uniform baseline ($\gamma = 0$ in Iteration 3), indicating a more refined and concentrated feature distribution. For instance, at Iteration 3, the configuration $\gamma = 0.7$ maintains a competitive LPIPS of 0.4614 while providing superior accuracy. This suggests that the proposed method effectively filters out marginal noise and focuses the model on high-fidelity semantic features, which is crucial for reliable compositional generation.

Table 4.5: Quantitative results: LPIPS scores across iterations and γ values.

Iteration	Random ($\gamma = 0$)	$\gamma = 0.3$	$\gamma = 0.5$	$\gamma = 0.7$	Balanced ($\gamma = 1$)
Iteration 1	0.5339	—	—	—	0.5596
Iteration 2	0.5276	0.5346	0.5269	0.5082	0.5333
Iteration 3	0.4461	0.4967	0.4675	0.4614	0.5117

4.4.2 Qualitative Analysis of Iterative Training on CelebA Dataset

To complement the quantitative findings, we perform a qualitative analysis using t-Distributed Stochastic Neighbor Embedding (t-SNE). This method allows us to project the high-dimensional

latent representations of the generated images into a two-dimensional plane, providing a visual assessment of the sample distribution across different sampling strategies.

Analysis of Iteration 2

As illustrated in Figure 4.2(a), the distribution of latent points in Iteration 2 reveals a clear correlation between the LPIPS diversity score and spatial dispersion. When the diversity score is higher, the t-SNE points are more spread out across the manifold, indicating a broader range of visual features. Conversely, lower diversity scores lead to more compact and dense clusters.

Specifically, in this stage, the configuration with $\gamma = 0.7$ (represented by the blue points) exhibits a noticeably more concentrated distribution compared to other variants. This observation aligns with its lower diversity score, suggesting that the model has begun to focus its generative capacity on a more refined set of semantic features for the unseen category.

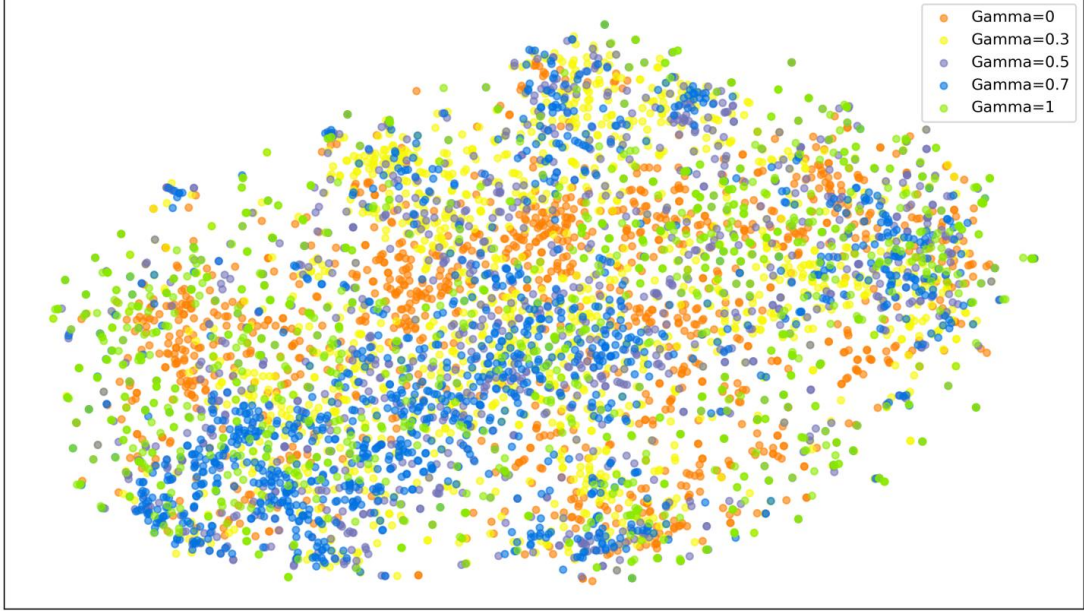
Analysis of Iteration 3

The contrast between sampling strategies becomes even more pronounced in Iteration 3, as shown in Figure 4.2(b). At this final stage of iterative refinement, we observe two extreme generative behaviors:

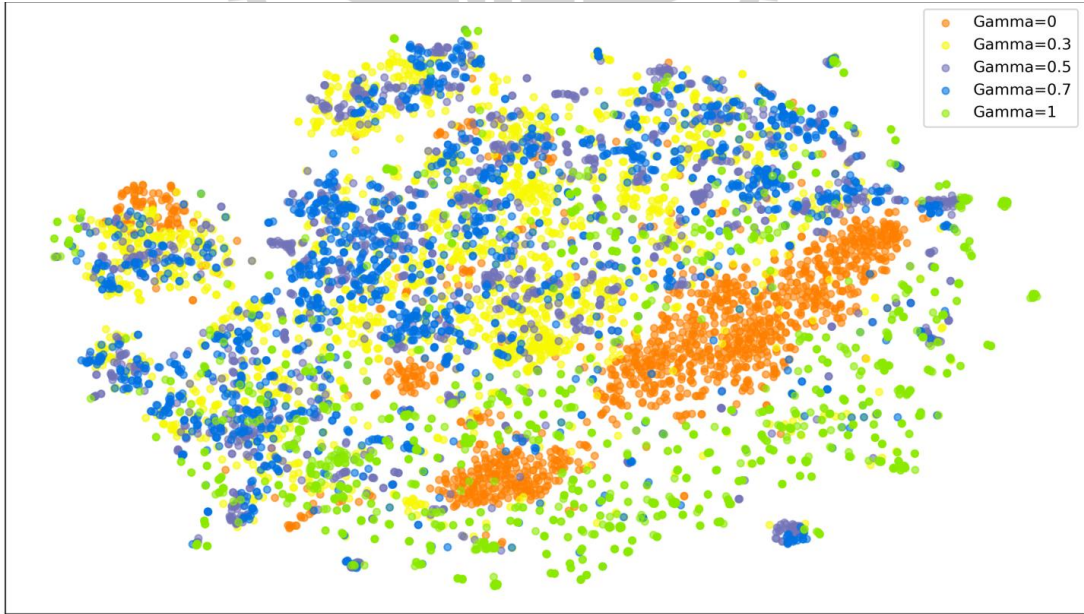
- **Low Diversity and Concentration ($\gamma = 0$):** The orange points representing uniform sampling ($\gamma = 0$) form a highly concentrated and dense cluster. This visual pattern corresponds to the lowest LPIPS score of 0.4461, confirming that a lower diversity score results in a more localized distribution in the latent space.
- **High Diversity and Dispersion ($\gamma = 1$):** In contrast, the green points representing balanced sampling ($\gamma = 1$) remain widely dispersed throughout the plot. This corresponds to the highest LPIPS score of 0.5117, demonstrating that the model maintains a significantly broader visual variety under this configuration.

The visual evidence from these t-SNE plots provides a strong validation of our proposed framework. The results confirm that by tuning the sampling parameter γ , we can effectively

control the focus of the generation space. Lower γ values (or lower LPIPS scores) result in more concentrated clusters, reflecting a model that has successfully identified and prioritized the core semantic features of the unseen combination, which is essential for achieving high accuracy in zero-shot compositional generation.



(a) t-SNE of Iteration 2



(b) t-SNE of Iteration 3

Figure 4.2: Visual analysis of latent space distributions. (a) In Iteration 2, $\gamma = 0.7$ shows increased concentration. (b) In Iteration 3, the extreme contrast between $\gamma = 0$ (Concentrated) and $\gamma = 1$ (Dispersed) demonstrates the relationship between LPIPS scores and visual diversity.

4.4.3 Results of Image Generation

Finally, we provide a qualitative assessment of the image generation results to visualize the model’s evolution throughout the iterative refinement process. Figure 4.3 presents a side-by-side comparison of the images generated for the target unseen category, “Gray Hair Female,” under the balanced sampling configuration ($\gamma = 1$).

As observed in the left panel of Figure 4.3, samples generated during **Iteration 1** exhibit high perceptual diversity, as reflected by the LPIPS score of 0.5596. However, the correctness ratio at this stage is relatively low at 14.92%. Many of these initial samples fail to accurately capture the specified compositional attributes, often manifesting as visual hallucinations where the hair color is inconsistent or the gender features are ambiguous. This indicates that without iterative refinement, a standard compositional DiT struggles to produce semantically reliable images for categories entirely absent from the training distribution.

Following the iterative human-in-the-loop (HITL) selection and retraining process, the results in **Iteration 3** (right panel) show a significant transformation in semantic fidelity. The correctness ratio experiences a substantial boost, reaching 34.16%. Although we observe a slight reduction in perceptual diversity, with the LPIPS score decreasing to 0.5117, this change represents a *strategic trade-off*. As the model is iteratively fine-tuned on human-curated correct samples, it effectively refines its internal generative space. By narrowing the focus toward high-fidelity semantic features, the model significantly improves its ability to synthesize accurate and realistic portraits that satisfy the specified compositional conditions.

These visual results confirm the effectiveness of our proposed framework. The synthesized gray-haired female images in the final iteration are not only visually plausible but also exhibit high semantic consistency. This demonstrates that by combining compositional Transformers with a curated iterative loop and balanced sampling, we can successfully bridge semantic data gaps and enable robust zero-shot image synthesis.

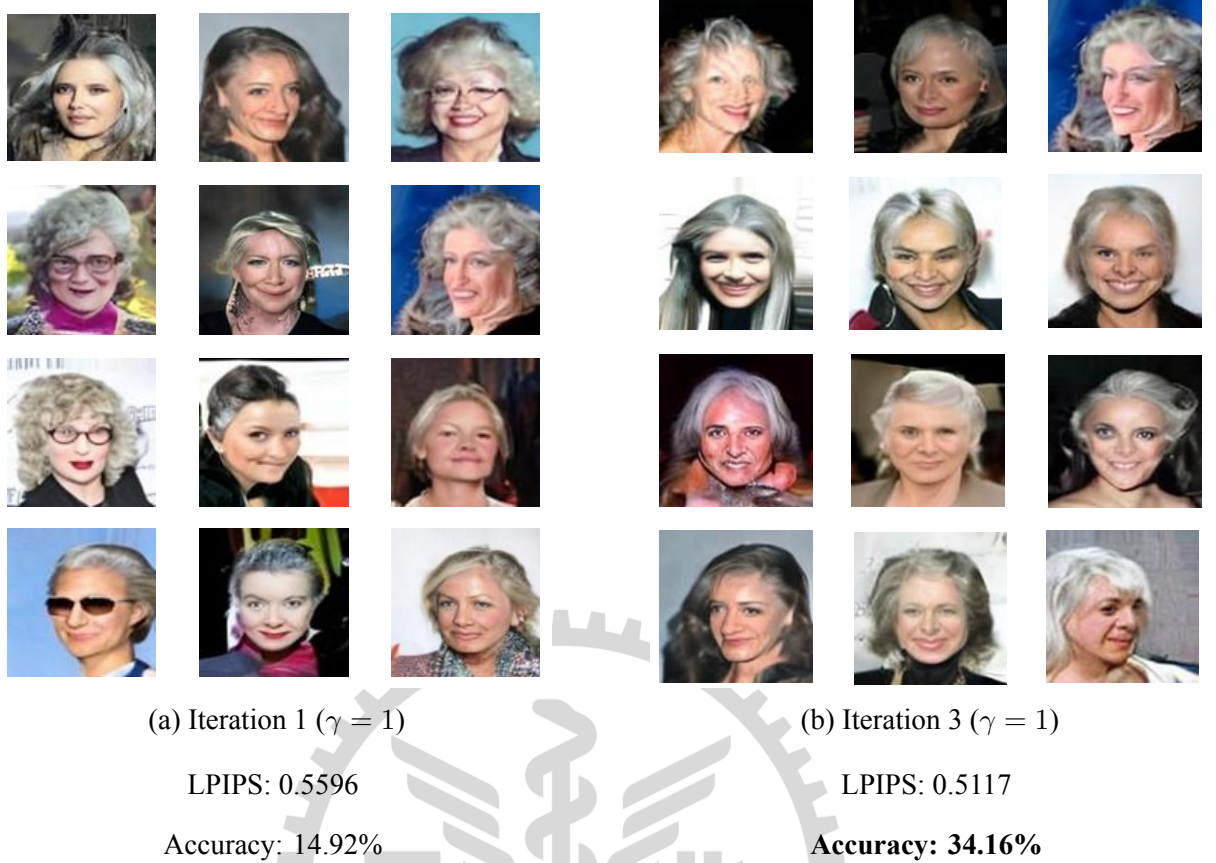


Figure 4.3: Visual comparison of generated “Gray Hair Female” samples. By strategically reducing marginal diversity through iterative refinement, the framework significantly enhances generation accuracy while maintaining sufficient visual variety for downstream utility.

4.4.4 Classifier-based Evaluation Results

To further verify the usefulness of the generated compositional images, we evaluate their impact on a downstream classification task under a missing-category setting. In this experiment, two binary classifiers are trained, namely an attribute classifier and an object classifier, to separately predict the corresponding attribute and object labels. A sample is considered correctly classified only when both classifiers make correct predictions. Therefore, the final composition accuracy is computed based on the intersection of the two classifier outputs.

The held-out test set consists of 1000 Gray_Hair Female images, while the corresponding compositional category is removed from the training set. This setting simulates a challenging scenario in which the classifier must generalize to an unseen composition. To examine whether the generated samples can compensate for the missing category, several sampling strategies are

Table 4.6: Classifier-based evaluation results under different sampling strategies across Iteration 2 and Iteration 3. The test set consists of 1000 Gray_Hair Female images. The final composition accuracy is computed by requiring both the attribute classifier and the object classifier to predict correctly.

Sampling Strategy	Iteration 2			Iteration 3		
	Attr.	Obj.	Comp.	Attr.	Obj.	Comp.
Baseline	230	949	205	—	—	—
Balanced Sampling	286	935	244	349	957	322
Random Sampling	280	951	245	234	946	203
Modified Balanced Sampling ($\gamma = 0.5$)	381	939	339	391	948	353
Modified Balanced Sampling ($\gamma = 0.3$)	361	929	310	397	927	350
Modified Balanced Sampling ($\gamma = 0.7$)	354	932	313	400	915	344

compared, including the baseline, balanced sampling, uniform sampling, and modified balanced sampling with different values of γ . In addition, the classifier performance is reported for both Iteration 2 and Iteration 3 in order to analyze the effect of iterative training on downstream classification performance.

Table 4.6 summarizes the classifier-based evaluation results under different sampling strategies. For each setting, the table reports the number of correct predictions made by the attribute classifier, the object classifier, and the final composition accuracy obtained from their intersection. The results show that the modified balanced sampling strategy consistently outperforms the baseline and the conventional sampling strategies in terms of composition accuracy.

Among all settings, the best performance is achieved by modified balanced sampling with $\gamma = 0.5$ at Iteration 3, which obtains a composition score of 353. This result indicates that the generated samples provide useful complementary information for improving downstream classification performance. Although some methods achieve relatively high scores on either the attribute classifier or the object classifier alone, the intersection-based composition accuracy shows that correctly recognizing both conditions simultaneously remains difficult without appropriate generated data. Overall, these results demonstrate that the proposed compositional image generation strategy is beneficial for improving classifier performance in the missing-category setting.

Chapter 5

Conclusion

In this work, we present a framework for compositional image generation using Diffusion Transformers (DiT). Different from traditional class-conditional DiT models that use a single atomic label, we introduce a compositional conditional training paradigm that factorizes class labels into independent attribute and object components, enabling zero-shot synthesis of unseen attribute—object combinations.

To improve generation reliability, we develop an iterative data refinement strategy with human-in-the-loop selection. In contrast to prior self-training methods that automatically use model predictions, our approach incorporates explicit human verification to ensure semantic correctness. This process is particularly relevant in scenarios with limited expert availability, where potential applications include dermatology (generated skin lesion images reviewed by dermatologists) and industrial wafer inspection (synthetic defective chip images reviewed by engineers for rare defect types).

Furthermore, we proposed a modified balanced sampling mechanism incorporating a tunable parameter, γ , to address the data imbalance introduced by iterative retraining. Unlike standard balanced or weighted sampling strategies that aim for total class re-balancing, our approach utilizes γ to regulate the re-balancing extent. This provides a flexible mechanism to manage the trade-off between increasing minority class exposure and avoiding excessive sample repetition. Experimental results demonstrate that the proposed sampling strategy effectively improves the overall performance.

Finally, our pipeline can be trained on consumer-grade hardware, providing a practical framework for compositional image generation with potential applicability to expert-driven, data-scarce domains.

References

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.
- [2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 8748–8763.
- [3] L. A. Gatys, A. S. Ecker, and M. Bethge, “Image style transfer using convolutional neural networks,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2414–2423.
- [4] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, and A. A. Efros, “Context encoders: Feature learning by inpainting,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 2536–2544.
- [5] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, A. Acosta, A. Aitken, A. Tejani, J. Totz, Z. Wang *et al.*, “Photo-realistic single image super-resolution using a generative adversarial network,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4681–4690.
- [6] D. P. Kingma and M. Welling, “Auto-encoding variational bayes,” *arXiv preprint arXiv:1312.6114*, 2013.
- [7] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, “Generative adversarial nets,” in *Advances in Neural Information Processing Systems*, vol. 27, 2014.

- [8] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, “Deep unsupervised learning using equilibrium thermodynamics,” in *International Conference on Machine Learning*. PMLR, 2015, pp. 2256–2265.
- [9] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.
- [10] M. Arjovsky, S. Chintala, and L. Bottou, “Wasserstein generative adversarial networks,” in *International Conference on Machine Learning*. PMLR, 2017, pp. 214–223.
- [11] P. Dhariwal and A. Nichol, “Diffusion models beat gans on image synthesis,” *Advances in neural information processing systems*, vol. 34, pp. 8780–8794, 2021.
- [12] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” in *International Conference on Medical Image Computing and Computer-assisted Intervention*. Springer, 2015, pp. 234–241.
- [13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020.
- [15] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 4195–4205.
- [16] Y. Atzmon, F. Kreuk, A. Feder, A. Polyak, Y. Gal, Y. Adi, and Y. Goldberg, “A learning-based approach to progressive hierarchy in image generation,” in *International Conference on Learning Representations*, 2020.

- [17] I. Misra, A. Gupta, and M. Hebert, “From red wine to red tomato: Compositional learning with context,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 1792–1801.
- [18] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, “Survey of hallucination in natural language generation,” *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.
- [19] D. Ferbach, Q. Bertrand, A. J. Bose, and G. Gidel, “Self-consuming generative models with curated data provably optimize human preferences,” *arXiv preprint arXiv:2407.09499*, 2024.
- [20] I. Shumailov, Z. Shumaylov, Y. Zhao, Y. Gal, N. Papernot, and R. Anderson, “Ai models make each other worse over time,” *Nature*, vol. 631, no. 8022, pp. 755–759, 2024.
- [21] Q. Bertrand, J. Bose, A. Duplessis, M. Jiralerspong, and G. Gidel, “On the stability of iterative retraining of generative models on their own data,” in *International Conference on Learning Representations*, vol. 2024, 2024, pp. 12 137–12 165.
- [22] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, “Class-balanced loss based on effective number of samples,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9268–9277.
- [23] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [24] G. Lemaître, F. Nogueira, and A. Christos, “Imbalanced-learn: A python toolbox to tackle the curriculum of imbalanced datasets in machine learning,” *Journal of Machine Learning Research*, vol. 18, no. 1, pp. 559–563, 2017.
- [25] Y. Shen, J. Gu, X. Tang, and B. Zhou, “Interpreting the latent space of gans for semantic face editing,” *CVPR*, 2020.

- [26] T. Nagarajan and K. Grauman, “Attributes as operators: factorizing unseen attribute-object compositions,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 169–185.
- [27] Y. Xian, S. Khare, Z. Akata, and B. Schiele, “Compositional zero-shot image generation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6495–6505.
- [28] S. L. Lai, P. C. Chen, and C. W. Ma, “Compositional conditional diffusion model,” in *2024 10th International Conference on Applied System Innovation (ICASI)*. IEEE, Apr. 2024, pp. 377–379.
- [29] Z. Liu, P. Luo, X. Wang, and X. Tang, “Deep learning face attributes in the wild,” in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 3730–3738.
- [30] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 586–595.